Moiré Plane Wave Expansion Model

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Simulating twisted van der Waals multilayer systems can be computationally very demanding, especially as the twist angle decreases due to the large number of atoms in the supercell. In this work, we introduce the energy- and height-dependent Moiré Plane Wave Expansion Model (MPWEM) which only requires the charge density data of the individual non-interacting two-dimensional material - that may be obtained by density functional theory (DFT) calculations - and uses its Fourier components to construct the charge density of the interacting multilayer system. With only a small number of intuitive parameters, such as the coupling strength between the component layers and screening, we can reconstruct the charge density of the interacting system with very good approximation. We illustrate the model with twisted bilayer graphene and $MoTe_2$/graphene moiré systems. This method can also be applied to incommensurate systems which lack translational symmetry as verified by experimental scanning tunneling microscopy in a variety of systems [1,2].

[1] M Le Ster et al., *Phys. Rev. B* **110**, 195418 (2024).

[2] W Ryś et al., *Nanoscale* **17**, 26835 (2025).